

ECE2301 Syllabus

ELCTRIC CIRCUITS I

FALL 2025

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TEXTBOOK: Fundamentals of Electric Circuits, 7th Edition. By Charles Alexander, Matthew Sadiku.

TIME/PLACE: T/Tr 12:00 – 1:20 PM / NURS 211

TEXTBOOK: 1. Sign in <https://connect.mheducation.com/class/h-erives-fall-2>
2. Subscribe

CATALOG DESCRIPTION: Introduction to systematic methodologies for the analysis of electric circuits in DC and AC steady state. Use of simulation tools for steady state circuit analysis. Can be taken concurrently with PHYS 2421 and MATH 2326.

PREREQUISITES: EE 1305, MATH 1312, PHYS 2421+ and MATH 2326+, each with a grade of C or better. (PHYS 2421 and MATH 2326 may be taken concurrently.)

COURSE LEARNING OUTCOMES: Students completing EE 2350 will be able to:

- Understand terminology used in conjunction with electric circuits of ideal circuit elements.
- Mathematically model electric systems using ideal resistive, inductive, and capacitive elements.
- Apply phasors and impedance transformations to the analysis of electric circuits.
- Apply various systematic methods (node, mesh, terminal equivalency, and circuit theorems).
- Apply various circuit analysis techniques to study circuits that include operational amplifiers.
- Apply various circuit analysis techniques to study energy and power in dc and ac circuits.
- Apply software tools to the analysis of electric circuits in steady state.

COURSE GRADING:

- Two Partial Exams 40%
- Homework (Connect) 30%
- Quizzes 20%
- Attendance 10%

Late work may have a late penalty of 30%.

GRADE DISTRIBUTION: Grades will be based on the standard scale

90% - 100% A

81% - 90% B

71% - 80% C

60% - 70% D

Below 60% F

TOPICS TO BE COVERED FROM THE TEXTBOOK: (the exact order, pages and/or sections and subsections will be listed in homework assignment handouts). See the Appendices for very useful review and reference materials!

Topics

Chapter 1: Circuit Variables

- a) Electrical Engineering
- b) Circuit Analysis: An Overview
- c) Power and Energy

Chapter 2: Circuit Elements

- a) Voltage and Current Sources
- b) Electrical Resistance (Ohm's Law)
- c) Kirchhoff's Laws

Chapter 3: Simple Resistive Circuits

- a) Resistors in Parallel and Series
- b) Voltage-Divider and Current-Divider
- c) Delta-to-Wye Equivalent Circuits

Chapter 4: Techniques of Circuit Analysis

- a) Node-Voltage Method and Special Cases
- b) Mesh-Current Method and Special Cases
- c) Source Transformations
- d) Thevening and Norton Equivalents
- e) Maximum Power Transfer
- f) Superposition

Exam I

Chapter 5: Operational Amplifier

- a) Op-Amp Terminals
- b) Terminal Voltages and Currents
- c) Inverting, Summing, Difference, and Noninverting-Amplifier Circuits

Chapter 6: Inductance, Capacitance, and Mutual Inductance

- a) The Inductor
- b) The Capacitor
- c) Mutual Inductance

Chapter 9: Sinusoidal Stead-State Analysis

- a) The Sinusoidal Source
- b) The Phasor

- c) Kirchhoff's Laws in the Frequency Domain
- d) Source Transformation and Thevening-Norton Equivalents
- e) Node-Voltage and Mesh-Current Methods
- f) The Transformer

Chapter 10: Sinusoidal Stead-State Power Calculations

- a) Instantaneous Power
- b) Average and Reactive Power
- c) The rms Value and Power Calculations
- d) Maximum Power Transfer

Exam II

ACADEMIC INTEGRITY:

Please review the statements below and UTEP's Web page on Policy on Academic Integrity at: <http://sa.utep.edu/osccr/academic-integrity/>. Academic dishonesty is prohibited and is considered a violation of the UTEP Handbook of Operating Procedures. It includes, but is not limited to, cheating, plagiarism, and collusion. Any act of academic dishonesty attempted by a UTEP student is unacceptable and will not be tolerated. Violations will be taken seriously and will be referred to the Dean of Students Office for possible disciplinary action. Students may be suspended or expelled from UTEP for such actions.

CENTER FOR ACCOMMODATIONS AND SUPPORT SERVICES (CASS):

If you have a disability and need classroom accommodations, please contact The Center for Accommodations and Support Services (CASS) at 747-5148, or by email to cass@utep.edu, or visit their office located in UTEP Union East, Room 106. For additional information, please visit the CASS website at www.sa.utep.edu/cass.

COVID-19 PRECAUTION STATEMENT

Please stay home if you have been diagnosed with COVID-19 or are experiencing COVID-19 symptoms. If you are feeling unwell, please let me know as soon as possible, so that we can work on appropriate accommodation. If you have tested positive for COVID-19, you are encouraged to report your results to covidaction@utep.edu, so that the Dean of Students Office can provide you with support and help with communication with your professors. The Student Health Center is equipped to provide COVID-19 testing.

The Center for Disease Control and Prevention recommends that people in areas of substantial or high COVID-19 transmission wear face masks when indoors in groups of people. The best way that Miners can take care of Miners is to get the vaccine. If you still need the vaccine, it is widely available in the El Paso area and will be available at no charge on campus during the first week of classes. For more information about the current rates, testing, and vaccinations, please visit epstrong.org